Effects of Non-Surgical Periodontal Treatment on Clinical Response, Serum Inflammatory Parameters, and Metabolic Control in Patients With Type 2 Diabetes: A Randomized Study

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Background: Scientific evidence on the effects of chronic periodontitis on diabetes mellitus remains inadequate and inconclusive. This intervention study is designed to evaluate the effects of periodontal treatment on clinical response, systemic inflammatory parameters, and metabolic control in patients with Type 2 diabetes.

Methods: A total of 134 patients were randomly allocated into two treatment groups and one control group. Treatment group 1 underwent non-surgical periodontal treatment at baseline and additional subgingival debridement at the 3-month follow-up. Patients in treatment group 2 received non-surgical periodontal treatment and supragingival prophylaxis at the 3-month follow-up, and those in the control group received no intervention throughout the study. All participants were reexamined at 1.5, 3, and 6 months after initial treatment. At each visit, clinical periodontal examinations were conducted and blood samples were taken to evaluate high-sensitivity C-reactive protein (hsCRP), tumor necrosis factor-α (TNF-α), glycated hemoglobin (HbA1c), fasting plasma glucose (FPG), and lipid profiles.

Results: Both treatment groups had a significantly lower hsCRP level after periodontal therapy ($P<0.05$). Although HbA1c declined significantly in treatment group 2 ($P<0.05$), the intergroup difference for HbA1c, FPG, TNF-α, and lipid profiles was not statistically significant after therapy ($P>0.05$).

Conclusions: Non-surgical periodontal treatment can effectively improve periodontal and circulating inflammatory status. Despite a lack of strong evidence, trends in some results support improved glycemic control after periodontal treatment in patients with diabetes. *J Periodontol 2012;83:435-443.*

KEY WORDS

C-reactive protein; diabetes mellitus; periodontitis; tumor necrosis factor-alpha.

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Because of differing pathogenesis, chronic periodontitis and Type 2 diabetes mellitus were previously thought to be unrelated; the former is a localized infection attributable to oral bacteria, whereas the latter is a systemic metabolic disorder characterized by abnormal glucose tolerance and hyperglycemia. However, more recently, evidence has indicated that these two conditions have much in common.[1-7] For example, both have similar polygenic backgrounds[1] and a relatively widespread prevalence in the general population, with similar clinical risk factors, such as smoking, age, psychologic stress,[2] and neutrophil functional alterations.[3] Additionally, both can upregulate the systemic immune response[4] with some degree of immunoregulatory dysfunction. Indeed, cross-susceptibility between periodontitis and diabetes[5] suggests a close two-way connection between them.

Although most of the available evidence tends to suggest that diabetes adversely affects periodontal health, some emerging clues tentatively support adverse effects of periodontitis on diabetes.[6] One potential mechanism for the influence of periodontitis on diabetes status may be that local periodontal infection results in a systemic burden of

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inflammatory mediators that exacerbate the existing metabolic disorder in patients with diabetes.[7]

Regarding the effects of periodontal infection on glycemic control in diabetes, longitudinal interventional studies provide more direct evidence[8] than other research methods such as cross-sectional surveys,[9,10] longitudinal observation studies,[11,12] and animal experiments.[13,14] However, there is no consensus as to whether successful periodontal therapy improves glycemic control in patients with diabetes and periodontitis. Using heterogeneous study designs, some researchers have reported improvement in glycemic control after the successful control of periodontal infection,[15-26] whereas other investigators have not found such a beneficial effect.[27-32] As pointed out in a recent meta-analysis,[33] related scientific evidence has remained inadequate and inconclusive to date. Thus, more randomized treatment studies would be helpful in investigating the influence of periodontal infection on metabolic control in patients with Type 2 diabetes.

Our preliminary observational study[9] suggested that chronic periodontitis may be associated with glycemic levels in patients with Type 2 diabetes. Based on these results, this interventional study focused on evaluating the influence of chronic periodontitis on diabetes, and had two main objectives: 1) evaluate the effects of non-surgical periodontal treatment on clinical response, metabolic control (including glycemic and lipid metabolism), and systemic inflammatory levels in patients with Type 2 diabetes, and 2) explore the effectiveness of supragingival prophylaxis versus that of subgingival debridement for supportive periodontal care after the treatment of chronic periodontitis in Type 2 patients with diabetes.

MATERIALS AND METHODS

Patients

From November 2008 to October 2009, 134 patients with diabetes (68 males and 66 females; aged 38 to 81 years) participating in a cross-sectional investigation described previously,[9] were recruited for this longitudinal study. All the patients had a confirmed diagnosis of Type 2 diabetes mellitus[34] for >1 year, with no change in their diabetic treatment plan in the previous 2 months and with no major diabetic complication, such as coronary heart disease. To be eligible for inclusion, participants had to have a clinical diagnosis of chronic periodontitis according to the American Academy of Periodontology criteria, with a ≥1 mm mean clinical attachment loss ([AL]; including slight, moderate, and severe periodontitis),[35] with ≥16 teeth. Exclusion criteria included: 1) the presence of a systemic disease other than diabetes that could influence the course of the periodontal disease; 2) systemic antibiotic administration within the previous 3 months; 3) pregnancy or lactation; or 4) refusal to provide written informed consent. Patients were also excluded if they had an active infection other than periodontitis or had received periodontal treatment in the previous 12 months.

Ethics approval was gained from the Medical Ethics Committee of Southern Medical University, Guangzhou, China, (Chinese Clinical Trial Registry Number ChiCTR-TRC-10001062) before implementation of the study. Written informed consent was obtained from all participants at the beginning of the study.

Demographic Data Collection

Demographic information was collected, including: 1) age; 2) sex; 3) history of diabetes; 4) use of medications; 5) smoking status; 6) alcohol consumption; and 7) physical exercise. Current smokers were defined as: patients who smoked ≥1 cigarette per day; those who had never smoked in their life (never smokers), and those who had given up smoking for >1 year (former smokers). Alcohol use was divided into drinking and not drinking, and diabetes mellitus treatment was classified as diet control, oral medication, or intravenous insulin use. The body mass index (BMI) was calculated by dividing the body weight (in kilograms) by the square of the height (in meters).

Study Design

This study is a randomized, parallel-control study conducted in Guangzhou, China. For allocation of the participants, a computer-generated list of random numbers was used, which was prepared by a statistician (YuXia Yan, Southern Medical University, Guangzhou, China) with no clinical involvement in the trial. All patients meeting eligibility criteria were numbered from 1 to 134 according to the recruitment sequence. Random assignment into three groups: treatment group 1, treatment group 2, and the control group, was then accomplished by an independent research nurse (BW) with a 1:1:1 allocation (Fig. 1). The allocation sequence was masked from the researcher (LC), who was responsible for enrolling and assessing participants named in sequentially numbered envelopes (1 to 134).

Patients in treatment group 1 received non-surgical periodontal treatment, which consisted of scaling and root planing under local anesthesia at baseline and additional subgingival debridement at the 3-month follow-up. Those in treatment group 2 underwent non-surgical periodontal treatment at the initial visit and only supragingival prophylaxis, with no intervention in deep periodontal pockets at 3 months. Those in the control group received no treatment measure or formal oral hygiene instructions until the end of the study. Non-surgical periodontal treatment was completed within 24 hours by an experienced periodontist (DX) without the administration of antibiotics or local antimicrobials at baseline, using standard rigid periodontal curets[‡] and ultrasonic instrumentation.[§]

‡ Gracey curets, Hu-Friedy, Chicago, IL.
§ Odontoson, Goof, Herlev, Denmark.

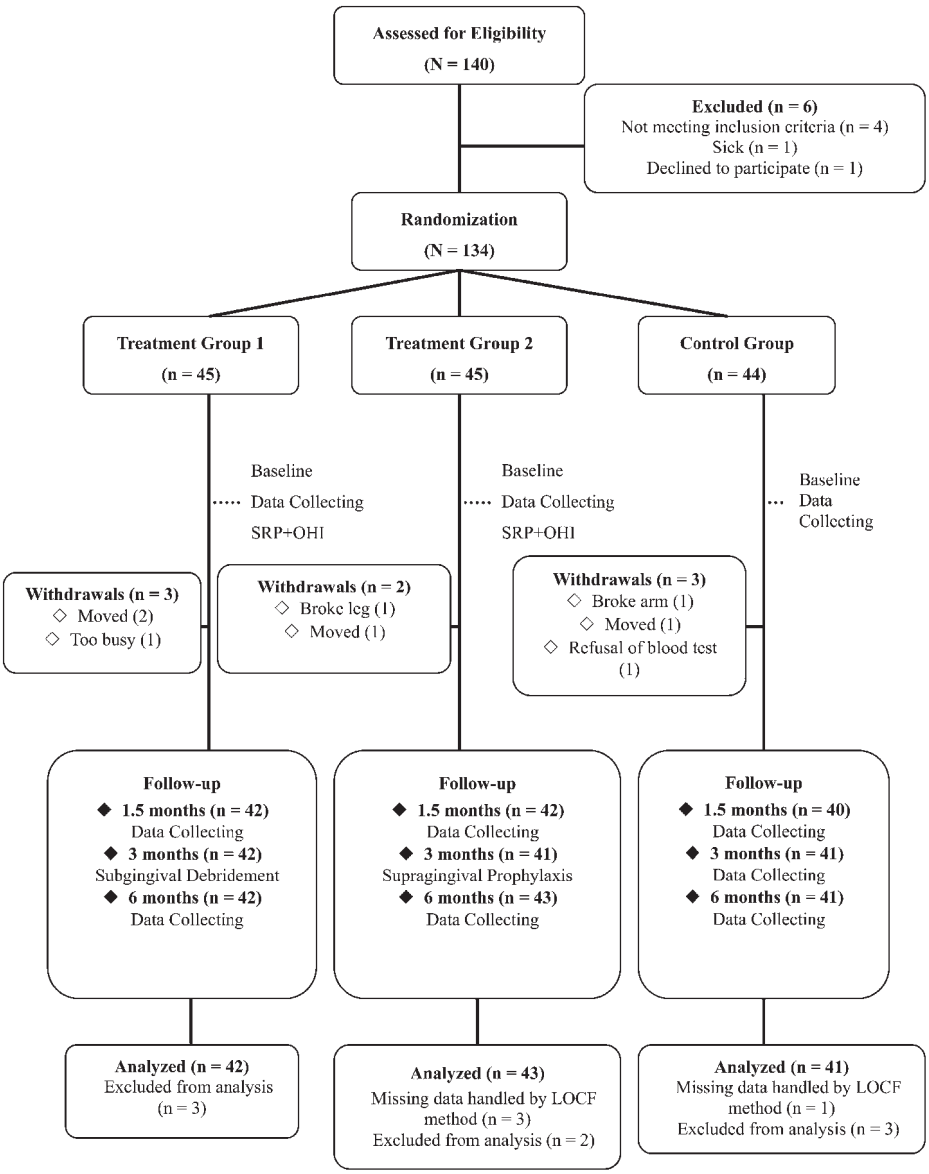

Figure 1.
Flow chart of study design and patient retention. SRP = scaling and root planing; OHI = oral hygiene instruction; LOCF = last observation carried forward.

Meanwhile, participants were instructed to continue with their medical treatment of diabetes mellitus, diet, and lifestyle without modification during the entire study period. Patients were then reexamined at 1.5, 3, and 6 months after completion of the initial periodontal therapy. At each visit, clinical periodontal examinations, blood tests, and immunologic studies were performed in all groups, with reinforcement of the oral hygiene instructions for patients in the two treatments groups.

Clinical Periodontal Examination

A previously calibrated examiner (LC; the intraexaminer κ value was 80% to 87%) completed a full-mouth assessment of periodontal conditions (except for third molars) at each visit, using a manual probe.‖ Clinical parameters at six sites per tooth (mesio-buccal, mid-buccal, disto-buccal, mesio-lingual, mid-lingual, disto-lingual) were recorded, including the plaque index (PI),[36] bleeding on probing (BOP),[37] probing depth (PD),[32] percentage of sites with PD ≤ 3 mm, PD = 4 to 5 mm, and PD ≥ 6 mm, gingival recession (GR), and clinical attachment level (CAL).[32]

Metabolic and Immunologic Study

At each visit, the primary efficacy measure (glycated hemoglobin [HbA1c]) was determined immediately by boronate-affinity chromatography, and secondary measures, including high-sensitivity C-reactive protein (hsCRP), fasting plasma glucose (FPG), triglycerides (TC), total cholesterol (TG), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), and tumor necrosis factor-α (TNF-α), were assessed by an independent and qualified commercial laboratory, as described in detail previously.[9]

Statistical Analyses

Assuming a standard deviation of 1.0%, enrollment of ≈53 patients per group should provide the study with a statistical power of 80% at a two-sided significance level of 0.05 based on an expected mean change of –0.50%, –0.30%, and 0.10% in HbA1c for treatment groups 1 and 2 and the control group, respectively (using statistical software).¶

For baseline information comparison, one-way analysis of variance (ANOVA) was used to assess continuous variables among the three groups, including age, duration of diabetes mellitus, BMI, and number of missing teeth. Pearson χ^2 test was performed to compare sex and regular physical exercise, and Fisher exact test was used to evaluate diabetes

‖ Williams manual probe, Hu-Friedy.
¶ PASS 2008, v.8.0.15, NCSS, Kaysville, UT.

Table 1.

Characteristics of the Study Groups at the First Visit (N = 126)

Variables	Treatment Group 1 (n = 42)	Treatment Group 2 (n = 43)	Control Group (n = 41)	*P*
Age (years; mean ± SD)	59.86 ± 9.48	57.91 ± 11.35	63.2 ± 8.51	0.052
Duration of diabetes mellitus (years; mean ± SD)	8.69 ± 5.25	6.93 ± 4.31	9.56 ± 6.02	0.066
BMI (mean ± SD)	24.46 ± 2.82	23.88 ± 3.56	23.51 ± 3.10	0.390
Number of missing teeth (mean ± SD)	3.60 ± 3.21	3.77 ± 3.67	4.46 ± 3.74	0.500
Male/female (n)	23/19	26/17	17/24	0.204
Smoking status (n)	NA	NA	NA	0.872
Current smoker	7	10	7	NA
Former smoker	1	1	0	NA
Never-smoker	34	32	34	NA
Diabetes mellitus treatment (n)	NA	NA	NA	0.574
Diet control	0	3	1	NA
Oral medication	38	35	36	NA
Insulin	4	5	4	NA
Alcohol consumption (n)	2	4	7	0.169
Regular physical exercise (n)	33	35	34	0.877

n = number of participants; NA = not applicable.

mellitus treatment, smoking status, and alcohol consumption (Table 1).

When homogeneity of the study groups at baseline was determined, a repeated-measures ANOVA (group factor: treatment group 1, treatment group 2, and control group; and time factor: initial visit, 1.5, 3, and 6 months) was used to analyze clinical, immunologic, and metabolic variables because these, or their log-transformed values (hsCRP, TNF-α), were normally distributed (Shapiro-Wilk test). Missing values were imputed using the last observation carried forward method.

Of 134 patients, eight withdrew and were excluded from statistical analysis. All statistical analyses were two tailed, used a significance level at 0.05, and were calculated using the software.[#]

RESULTS

Patient Retention

After applying the inclusion and exclusion criteria, 134 patients with diabetes met the recruitment criteria. Of these, 126 patients (treatment group 1, n = 42; treatment group 2, n = 43; control group, n = 41) (Fig. 1) were included in the analyses, and eight patients withdrew after randomization. All patients in treatment group 1 continued to the end of the maintenance program. In treatment group 2, one woman missed her 1.5-month follow-up, and two men missed their 3-month follow-ups as a result of scheduling conflicts, but each attended the subsequent visits. In the control group, one female patient missed her 1.5-month follow-up but completed the following ones. No patient reported any adverse effect, such as fever or acute onset of periodontal infection, during the study.

Demographic Data

Patients were homogeneous with respect to demographic variables. No significant difference was found in the baseline data, such as age, duration of diabetes mellitus, BMI, number of missing teeth, sex, smoking status, diabetes mellitus treatment, regular physical exercise, or drinking alcohol among groups ($P > 0.05$; Table 1).

Clinical Periodontal Response

No significant difference among groups was evident in almost any of the periodontal parameters at baseline ($P > 0.05$), including PI, BOP, mean PD, mean CAL,

SPSS v.13.0, IBM, Chicago, IL.

Table 2.

Periodontal Clinical Parameters in the Three Groups Over the Study Period (mean ± SD)

Variables	Groups	Baseline	1.5 Months	3 Months	6 Months
PI	Treatment group 1	0.94 ± 0.42	0.47 ± 0.31*†	0.40 ± 0.25*†	0.45 ± 0.27*†
	Treatment group 2	0.74 ± 0.38	0.45 ± 0.30*†	0.42 ± 0.31*†	0.40 ± 0.28*†
	Control group	0.93 ± 0.56	0.78 ± 0.42*	0.74 ± 0.40*	0.75 ± 0.38*
BOP (%)	Treatment group 1	37.74 ± 20.04	16.90 ± 12.98*†	13.36 ± 10.78*†	13.86 ± 10.95*†
	Treatment group 2	32.42 ± 16.63	14.16 ± 11.10*†	12.13 ± 8.24*†	12.02 ± 8.99*†
	Control group	34.01 ± 18.91	29.76 ± 14.85*	28.53 ± 14.42*	28.37 ± 13.50*
Sites with PD = 4 to 5 mm (%)	Treatment group 1	12.93 ± 10.48	8.28 ± 8.36*	7.20 ± 7.86*	6.18 ± 7.42*
	Treatment group 2	10.44 ± 10.33	5.78 ± 6.56*	6.09 ± 7.29*	5.53 ± 7.09*
	Control group	9.44 ± 9.58	8.48 ± 8.53	8.17 ± 8.47	8.46 ± 9.10
Sites with PD ≥6 mm (%)	Treatment group 1	4.74 ± 7.35	1.98 ± 3.69*	2.10 ± 4.23*	2.76 ± 8.99*
	Treatment group 2	4.20 ± 6.97	1.47 ± 3.01*	1.22 ± 2.45*	1.24 ± 2.91*
	Control group	2.22 ± 3.04	1.87 ± 2.55	2.29 ± 5.01	3.04 ± 7.19
Mean PD (mm)	Treatment group 1	2.66 ± 0.68	2.32 ± 0.58*	2.27 ± 0.50*	2.18 ± 0.48*†
	Treatment group 2	2.57 ± 0.66	2.20 ± 0.41*	2.20 ± 0.39*	2.10 ± 0.39*†
	Control group	2.47 ± 0.57	2.40 ± 0.49	2.38 ± 0.47	2.42 ± 0.50
Mean CAL (mm)	Treatment group 1	3.57 ± 1.31	3.26 ± 1.26*	3.28 ± 1.25*	3.20 ± 1.23*
	Treatment group 2	2.95 ± 1.21	2.56 ± 1.15*†	2.55 ± 1.15*†	2.55 ± 1.16*†
	Control group	3.37 ± 1.24	3.27 ± 1.22	3.29 ± 1.23	3.41 ± 1.23

* Statistically significantly different from baseline (P <0.05).
† Statistically significantly different from the control group (P <0.05).

percentage of sites with PD = 4 to 5 mm, and PD ≥ 6 mm (Table 2). Only mean GR in treatment group 2 was significantly lower than that in treatment group 1 and the control group at initial visit (data not shown; P = 0.037).

In both treatment groups, PI and BOP decreased significantly at 1.5 months after the non-surgical periodontal treatment, and these improvements were maintained throughout the study period. In the control group, PI and BOP declined significantly at 1.5 months compared to baseline values (P<0.05). Comparisons among groups showed that PI and BOP in both intervention groups were significantly lower than those in the control group at 1.5 months (P <0.05), and these significant differences remained until the end of the study. No significant difference was found in PI or BOP between the two therapeutic groups (P>0.05).

The percentage of sites with PD = 4 to 5 mm and PD ≥6 mm declined significantly over time in both treatment groups (P<0.05), whereas it did not change significantly in the control group throughout the study period (P>0.05). However, there were no statistically significant difference in these variables among the three groups at each time point.

Significant differences existed between the intervention groups and the control group (P <0.05) but not between the treatment groups (P>0.05).

A significant decline in mean CAL was observed in both therapeutic groups at 1.5 months after initial non-surgical periodontal treatment (P <0.05), and the improvement remained stable thereafter. In contrast, no significant difference was found in mean CAL at 6 months compared to baseline values in the control group (P >0.05). Comparisons among groups showed the mean CAL in treatment group 2 was significantly lower than that in the other groups at follow-up (P<0.05).

Metabolic Response

At baseline, the three groups had similar mean values of HbA1c, FPG, TC, TG, HDL-C, and LDL-C (P>0.05).

A significant reduction in HbA1c was seen at 6 months in treatment group 2 (P<0.05) but not in treatment group 1 or the control group (P>0.05) (Table 3). Comparisons among the three groups showed no significant difference in HbA1c at baseline or 1.5, 3, or 6 months (P>0.05).

A significantly lower FPG was observed in treatment group 1 at 6 months (P<0.05) but not in treatment group 2 or the control group (P >0.05). The three groups did not differ significantly in FPG at any follow-up visit (P>0.05).

Lipid metabolic levels, including TC, TG, HDL-C, and LDL-C, in all three groups had a tendency to decline during the whole study period, but comparisons among the three groups showed no significant difference in these variables at any time point (P>0.05).

Table 3.

Metabolic and Immunologic Markers in the Three Groups Over the Study Period (mean ± SD)

Variables	Groups	Baseline	1.5 Months	3 Months	6 Months
HbA1c (%)	Treatment group 1	7.31 ± 1.23	7.21 ± 1.55	7.30 ± 1.50	7.09 ± 1.34
	Treatment group 2	7.29 ± 1.55	7.24 ± 1.33	7.43 ± 1.53	6.87 ± 1.12*
	Control group	7.25 ± 1.49	7.39 ± 1.54	7.59 ± 1.54	7.38 ± 1.57
FPG (mmol/L)	Treatment group 1	7.65 ± 1.72	7.44 ± 1.83	7.41 ± 1.66	6.92 ± 1.80*
	Treatment group 2	7.68 ± 2.05	7.99 ± 2.17	7.93 ± 2.08	7.48 ± 1.75
	Control group	7.96 ± 2.46	8.49 ± 3.10	8.30 ± 2.67	8.16 ± 2.80
hsCRP (mg/L)	Treatment group 1	3.21 ± 4.45	2.38 ± 2.88	1.48 ± 1.00*	1.58 ± 1.31*†
	Treatment group 2	3.09 ± 4.64	1.88 ± 2.83	1.96 ± 1.98	1.53 ± 1.27*†
	Control group	2.81 ± 4.05	2.36 ± 1.94	2.24 ± 2.06	3.16 ± 5.45
TNF-α (pg/mL)	Treatment group 1	11.19 ± 3.97	10.12 ± 3.30	11.37 ± 5.18	10.78 ± 2.97
	Treatment group 2	12.43 ± 15.55	10.68 ± 3.63	10.37 ± 4.39	11.03 ± 3.77
	Control group	10.52 ± 3.37	10.19 ± 2.81	11.33 ± 3.75	10.98 ± 2.42
TC (mmol/L)	Treatment group 1	2.09 ± 1.16	1.89 ± 1.14	1.95 ± 1.10	1.90 ± 1.11
	Treatment group 2	2.63 ± 1.32	2.29 ± 1.28	2.20 ± 1.33	2.15 ± 1.93
	Control group	2.35 ± 1.78	2.26 ± 1.61	2.30 ± 2.16	2.25 ± 1.98
TG (mmol/L)	Treatment group 1	5.91 ± 1.20	5.77 ± 1.31	5.75 ± 1.14	5.64 ± 1.37
	Treatment group 2	6.02 ± 1.57	5.65 ± 1.35*	5.61 ± 1.41*	5.26 ± 1.41*
	Control group	6.37 ± 1.87	6.03 ± 2.03	5.94 ± 1.22	5.81 ± 1.61
HDL-C (mmol/L)	Treatment group 1	1.30 ± 0.39	1.31 ± 0.38	1.27 ± 0.38	1.19 ± 0.34*
	Treatment group 2	1.31 ± 0.46	1.25 ± 0.40	1.23 ± 0.39	1.14 ± 0.39*
	Control group	1.44 ± 0.53	1.41 ± 0.60	1.36 ± 0.49	1.26 ± 0.50*
LDL-C (mmol/L)	Treatment group 1	3.64 ± 0.99	3.48 ± 0.96	3.49 ± 0.85	3.29 ± 0.88*
	Treatment group 2	3.50 ± 1.30	3.34 ± 1.17	3.37 ± 1.30	3.04 ± 1.23*
	Control group	3.79 ± 1.48	3.57 ± 1.66	3.50 ± 1.17	3.25 ± 1.27*

* Statistically significantly different from baseline ($P<0.05$).
† Statistically significantly different from the control group ($P<0.05$).

Immunologic Response

The three groups did not differ in average values for hsCRP or TNF-α levels at baseline ($P>0.05$) (Table 3).

Serum hsCRP in both treatment groups declined significantly over time ($P<0.05$), whereas hsCRP in the control group showed no significant difference ($P>0.05$). Compared to the control group, both therapeutic groups had significantly lower hsCRP at 6 months ($P<0.05$), with no significant difference between them ($P>0.05$).

During the entire study period, the serum TNF-α fluctuation showed no statistically significant difference in any group ($P>0.05$). Also, no significant intergroup difference was found at any time point ($P>0.05$).

DISCUSSION

Research has shown a reciprocal association between periodontal disease and diabetes mellitus, specifically, patients with diabetes are more likely to develop periodontal disease than are their healthy counterparts,[6] and periodontal infection may have an adverse effect on glycemic control and the incidence of diabetes complications.[8] Accordingly, it has been suggested that periodontal therapy, which decreases the intraoral bacterial bio-burden and reduces periodontitis-induced bacteremia/endotoxemia, may have a positive impact on systemic inflammatory status and metabolic control in patients with diabetes.[8]

To prevent the recurrence and progression of periodontal infection and thus maintain improved metabolic control obtained after periodontal therapy,[19] supportive periodontal care should be performed at regular intervals after initial periodontal treatment. However, because of a lack of clinical evidence, recommendations for the optimal regimen for periodontal maintenance[38] and the recall intervals vary widely between countries and dental health care systems.[39] Furthermore, a prospective study[40] called

into question the value of performing repeated subgingival debridement at 3-month intervals for patients with periodontitis. In such a context, this prospective, randomized, controlled-treatment study focuses on evaluating the potential value of supragingival prophylaxis in contrast with subgingival debridement after non-surgical periodontal treatment on clinical response, systemic inflammatory levels, and metabolic control in patients with diabetes with periodontitis.

Treatment of group 1 differed from that of group 2 in the periodontal maintenance regimen (subgingival debridement versus supragingival prophylaxis, respectively) at the 3-month visit. This study shows that the two regimens were comparable with respect to such periodontal variables as PI, BOP, mean PD, percentage of sites with PD = 4 to 5 mm, and PD ≥ 6 mm, indicating that repeated subgingival debridement at 3 months after initial treatment did not lead to a better clinical outcome. These results are consistent with a meta-analysis[38] and a previous clinical study[40] that failed to show that repeated subgingival scaling at 3-month intervals was superior to coronal scaling during a 1-year maintenance program. With regard to mean CAL, contrary to our expectations, treatment group 2 showed a better clinical outcome than group 1 at each follow-up, which was not caused by different supportive periodontal care measures, because significant difference between the groups had been observed even before implementation of periodontal maintenance measures. This outcome might be explained by a different mean GR (1.60 ± 0.80 mm for group 1 and 1.25 ± 0.67 mm for group 2) in the two groups at baseline. We suggest that patients with equal mean PD and lower mean GR had a greater surface area of inflamed pocket epithelium and thus had a better clinical response to non-surgical treatment. In conclusion, both supragingival prophylaxis and subgingival debridement were effective in the control of periodontal inflammation in diabetes after non-surgical periodontal treatment.

It has been suggested that periodontal treatment may improve glycemic control through a reduction in CRP, a significant risk indicator for both the development of cardiovascular disease and the progression of diabetes in patients with diabetes and periodontal disease.[19] In this study, in association with significant improvements in periodontal measures, the hsCRP level declined markedly over time in both intervention groups, suggesting that non-surgical periodontal treatment contributed to improved systemic inflammatory status, which, in turn, helped to lower the risk for the development of microvascular complications in patients with diabetes.[41] Furthermore, the two regimens (supragingival prophylaxis and subgingival debridement) of supportive periodontal care were equally effective in lowering hsCRP level; no significant difference was found between them. These outcomes also suggest that the general population of patients with diabetes and periodontitis would benefit from periodontal treatment with respect to hsCRP level because this intervention study was conducted with patients of both sexes, all ages, differing levels of glycemic control, and a broad spectrum of periodontal infection severity, from slight to moderate and severe periodontitis. Thus, it is recommended that patients with diabetes and periodontitis receive periodontal treatment to reduce the risk of diabetes complications through reduction of hsCRP.

Unexpectedly, changes in HbA1c did not parallel those in clinical periodontal variables or CRP. This might be attributed to the fact that HbA1c, which reflects the average blood glucose level of the previous 90 days, lags behind changes in inflammatory markers. In contrast to a recent study[42] that found improved glycemic control after periodontal treatment in diabetic patients with moderate-to-severe periodontitis, the current trial did not show a statistically significant intergroup difference in HbA1c or FPG at the 6-month follow-up. Such an outcome does not provide strong evidence in support of the concept that non-surgical periodontal treatment can improve metabolic control in an overall population of patients with Type 2 diabetes and chronic periodontitis. A possible explanation for this limited effect of therapeutic periodontal improvement on diabetic metabolic control may be that patients enrolled in this trial did not have severe periodontal disease (mean values for percentage of sites with PD ≥4 mm ranged from 11.68% to 17.68% at the initial visit). However, some results do suggest glycemic improvement after non-surgical periodontal treatment. For example, HbA1c declined significantly in treatment group 2 at 6 months, although not in group 1, which may be explained by the better clinical periodontal response in group 2.

An important limitation of this study is simply that it was underpowered. Larger controlled trials are needed to clarify the effects of periodontal therapy on glycemic control in diabetes patients and to determine which patients, in terms of extent or severity of periodontitis, can be expected to benefit most and show a significant decrease in HbA1c after periodontal treatment.

To date, the role of TNF-α in patients with diabetes and periodontitis remains controversial. Some intervention studies have reported that periodontal treatment could improve glycemic control in diabetes by reducing peripheral TNF-α concentrations,[16,18] whereas other studies have failed to find such a reduction in serum TNF-α level after periodontal therapy in patients with diabetes or systemically healthy populations.[15,22,25,43-45] Results of this longitudinal study

showed that TNF-α did not differ by time point in each group, nor did it differ among the three groups at any time point, indicating that non-surgical periodontal therapy had no significant impact on serum TNF-α concentrations in these patients with Type 2 diabetes. These findings, combined with our previous cross-sectional investigation showing little impact of the severity of periodontal inflammation on serum TNF-α levels,[9] suggest that increased circulating TNF-α concentrations, caused by periodontal infection, are too minor to play a major role in any adverse effect of periodontitis on diabetes.

CONCLUSIONS

The results of this randomized, controlled clinical trial suggest that non-surgical periodontal treatment can effectively control periodontal inflammation and reduce serum hsCRP level in patients with Type 2 diabetes and periodontitis. Despite a lack of strong evidence, trends in some results show evidence of somewhat improved glycemic control after periodontal treatment in patients with diabetes.

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